Audrey Houghton (she/her)

github.com/audreymhoughton | linkedin.com/in/audreyhoughton | audreymhoughton@gmail.com
+1-612-462-1481 | Google Scholar

Technical Skills

Programming: Python, Bash, UNIX/Linux, Git/GitHub, LaTeX, Docker, Singularity
Infrastructure & Data: AWS (S3), Ceph, SLURM, Kubernetes, SQL, ETL pipelines, batch processing, data validation, Globus, Grafana, Prometheus
Statistics & Modeling: logistic regression, predictive modeling, LightGBM, calibration (Platt/isotonic), repeated stratified cross-validation, bootstrap/permutation inference, FDR correction (Benjamini-Hochberg), SHAP interpretability, ROC-AUC/PR-AUC/Brier metrics, causal inference & discovery, GIS analysis, QGIS visualization, EHR integration, geocoding/FIPS recovery
Imaging & ML: neuroimaging/MRI segmentation, astrophysics photometric reduction, time-domain imaging, deep learning inference workloads and compute planning (e.g., nnU-Net)
Productivity & Collaboration: Jira, Linear, Kanban, Agile/Scrum, SOP documentation, Slack, Microsoft Teams, Google Workspace, Box, Salesforce API automation, compliant web scraping
Additional Exposure: Matlab, R, IDL, JavaScript/Node.js, Go, Ansible/AWX, RedHat, ReadTheDocs, Google Apps Script, GCP, Terraform, Datadog, CI/CD (GitHub Actions, CircleCI)

Experience

University of Minnesota – Department of Psychiatry & Behavioral Sciences *Dec 2025 - Present*
Research Software Engineer III Minneapolis, MN (Remote)

- Develop and maintain Python analysis pipelines for GIS-linked electronic health record data in psychiatric research
- Build geospatial feature-engineering workflows in Python/Jupyter for county- and tract-level overlays, geocoding quality control, and FIPS recovery from external APIs
- Build publication-grade clinical prediction workflows with LightGBM, repeated stratified cross-validation, calibration selection, and bootstrap/permutation robustness checks
- Lead statistical result interpretation and methods reporting for manuscripts and grant submissions across interdisciplinary teams

CoreWeave, Inc. *Apr 2024 - Sept 2024*
Fleet Operations Engineer Remote

- Provisioned and operated thousands of NVIDIA GPU nodes in Kubernetes using Ansible, AWX, Grafana, and Go, achieving 1-2 week onboarding and 6+ months average production uptime
- Diagnosed and resolved hardware/software failures across the node lifecycle, recovering approximately $250,000 in value per repaired node
- Partnered with 10+ teams—including networking, hardware, operations, and support—to streamline troubleshooting, accelerate incident resolution, and document SOPs and post-mortems, improving efficiency and reliability
- Identified points in the node lifecycle where nodes were lost, implementing solutions that saved the company millions in revenue

Masonic Institute of the Developing Brain *Jan 2021 - Apr 2024*
Research Software Engineer II (promoted to III) Minneapolis, MN (Remote)

- Developed Python wrappers leveraging AWS S3 storage and SLURM optimization to accelerate image processing up to 12x
- Engineered a containerized infant MRI segmentation application with deep learning models, achieving 600x speed-ups and +53% accuracy
- Built ETL-based pipelines to transfer and process petabytes of imaging data 10x faster than prior industry benchmarks, incorporating statistical error analysis and validation to ensure reliability and reproducibility
- Authored comprehensive documentation and curriculum, setting standards for reproducible neuroimaging data processing

Calyxt *Oct 2019 - Aug 2020*
Laboratory Assistant & Data Specialist Roseville, MN

- Developed Python scripts for laboratory data tracking, quality control, and analysis reporting

The MINERVA Project *Jan 2015 - May 2017*
Research Assistant Missoula, MT

- Developed a Python-based photometric data reduction pipeline for detecting transiting exoplanets from time-domain imaging data, including calibration, aperture photometry, and light curve analysis

Rutgers University Astrophysics Department *May 2016 - Aug 2016*
Research Assistant New Brunswick, NJ

- Refactored IDL scripts into Python
- Built galactic simulation models using parameter optimization

Products

Sales Summary Pipeline 2025
Independent Automation Project [Proprietary]

- Built a proof-of-concept pipeline integrating Salesforce logging, CSV rotation, and Microsoft Teams alerts with compliant web scraping (robots.txt, throttling, retries)

CoreWeave Internal Documentation 2024
CoreWeave, Inc. [Proprietary]

- Initiated and built company-wide documentation from the ground up, replacing scattered one-off documents with an organized, centralized Notion knowledge base covering terminology, SOPs, and team workflows

SLURM Wrappers 2021 - 2024
High-Performance Computing S3 and Disk SLURM-Compliant Wrappers github.com/DCAN-Labs/SLURM_wrappers

- Bash/Python wrappers adaptable to any SLURM-based data pipeline

ABCD Study Collection 3165 Data Processing 2021 - 2024
NIH Data Archive (NDA) collection3165.readthedocs.io

- Community dataset enabling access to petabytes of processed/unprocessed MRI data under BIDS standards

DCAN Labs Informational Guide 2022 - 2024
Minnesota Supercomputing Institute Data Engineering Documentation dcan-labs-informational-guide.readthedocs.io

- Comprehensive, continually-supported documentation of usage of the Minnesota Supercomputing Institute to process, analyze, and transfer MRI data

CABINET 2022 - 2023
Container-Linking Wrapper github.com/DCAN-Labs/CABINET

- Python-based universal container-linking wrapper employing FAIR principles

BIBSNet 2022 - 2023
Infant MRI Brain Segmentation Application github.com/DCAN-Labs/BIBSnet

- Containerized infant MRI application; achieved 600x faster processing and +53% accuracy

Education

University of Montana - Missoula, MT Aug 2013 - May 2017
Bachelor of Arts in Physics umt.edu/physics-astronomy

University of Minnesota 2023
Data Management Series; Project Management Certificate Coursework